

Potential Energy Diagrams — 5E Lesson Plan

Grade Level

11th Grade Regents Chemistry

Time Required

Approximately 50 minutes (1 class period)

I would spend two full class periods on this lesson

Performance Expectations (NYSSLS-Aligned)

HS-PS1-4: Develop a model to illustrate that the release or absorption of energy from a chemical reaction system depends upon the changes in total bond energy.

HS-PS3-1: Create a computational model to calculate the change in energy of one component in a system when the change in energy of the other component(s) and energy flows in and out of the system are known.

Learning Objectives

1. Interpret potential energy diagrams for endothermic and exothermic reactions.
2. Identify activation energy, activated complex (transition state), and heat of reaction (ΔH) on a diagram.
3. Explain energy flow in terms of bond breaking and bond forming.
4. Relate energy diagrams to real-world chemical processes.

Materials

Whiteboard or display, colored markers, blank potential energy diagram handouts, calcium chloride, ammonium nitrate, thermometers, beakers, graph paper or student devices, and exit ticket slips.

ENGAGE (5–7 minutes)

Begin with instant hot and cold packs to spark curiosity. Perform quick demonstrations showing exothermic ($\text{CaCl}_2 + \text{H}_2\text{O}$) and endothermic ($\text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$) reactions. Ask students to predict energy flow and introduce the concept of potential energy diagrams.

When chat GPT makes PDF files it screws up some symbols

instead of a demo have students conduct these themselves.

EXPLORE (10–12 minutes)

Students graph temperature data for both reactions on a basic energy vs. reaction progress graph. They infer which reactions absorb or release energy and discuss activation energy as a 'hill' on the graph.

EXPLAIN (15 minutes)

Teacher introduces formal diagrams for exothermic and endothermic reactions, labeling reactants, products, activated complex, activation energy (E_a), and heat of reaction (ΔH). Use colors and analogies to explain bond breaking/forming and net energy change.

this would be initial and final temp data.

ELABORATE (10 minutes)

Students match real-world examples (e.g., combustion, photosynthesis) to energy diagrams and justify their choices. Optionally, introduce catalysis by showing a reduced activation energy curve and discuss its impact on reaction rate.

Use the analogy of a person on a bike going over a hill (activation energy)

print matching cards on cardstock

at this point introducing a catalyst would be too much added complexity

EVALUATE (6–8 minutes)

Students complete an exit ticket drawing a labeled potential energy diagram and writing one sentence explaining the energy change. Discussion questions reinforce concepts such as identifying exo-/endothermic reactions from diagrams.

Differentiation & Support

- Visual learners: color-coded diagrams and posted anchor chart.
- ELL students: word bank with visuals for key terms.
- **Advanced learners:** compare catalyzed and uncatalyzed pathways or calculate ΔH from given energy values.

Closure / Real-World Connection

Discuss how potential energy diagrams help chemists design efficient reactions, such as in batteries or pharmaceuticals.

Homework / Extension

Find a real-world example of an endothermic or exothermic reaction, classify it, and sketch its potential energy diagram.